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Projector arrangement for a vehicle

Abstract

A projector arrangement for a vehicle includes an attachment mechanism that selectively couples with a rear door of the vehicle. The projector arrangement further includes a video projector coupled with the attachment mechanism. The video projector is configured to project image data in a direction toward the vehicle when the rear door is in an open position. A portable container selectively houses the attachment mechanism and the video projector.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to a projector arrangement for a vehicle and, more particularly, to a portable projector arrangement that selectively couples with a rear door of a vehicle.

BACKGROUND OF THE DISCLOSURE

(2) Conventional vehicles typically include infotainment systems and/or entertainment accessories. These infotainment and entertainment systems have advanced over time to allow consumers to experience media in various ways. For example, conventional vehicles may include flip down display screens for displaying video.

(3) Typically, consumers are required to install these aftermarket display assemblies or purchase separate and incompatible display systems for a given vehicle. In addition, if a projection device is implemented in a vehicle, structures of the vehicle cabin may be required to be rearranged to limit obstruction of the video projection.

SUMMARY OF THE DISCLOSURE

(4) According to a first aspect of the present disclosure, a projector arrangement for a vehicle includes an attachment mechanism selectively coupling with a rear door of the vehicle. The projector arrangement further includes a video projector coupled with the attachment mechanism. The video projector is configured to project image data in a direction toward the vehicle when the rear door is in an open position.

(5) Embodiments of the first aspect of the disclosure can include anyone or a combination of the following features: the attachment mechanism including at least one clip configured to engage a handle of the rear door; the attachment mechanism including at least one elongate member extending between the video projector and the at least one clip; the at least one clip including a first clip corresponding to a first elongate member and the second clip corresponding to a second elongate member, the first and second elongate members laterally spacing the video projector from the handle when the attachment mechanism is coupled to the rear door; the handle and the clip forming a snap-fit connection; a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in a closed position; the at least one bracket including a plurality of brackets, and the plurality of brackets configured to secure the screen in a planar configuration; each of the plurality of brackets including a first hook configured to engage a wall of the vehicle; the screen defining a plurality of holes disposed proximate to a periphery of the screen, and each of the plurality of brackets including a second hook opposite the first hook configured to engage one of the plurality of holes; a portable container configured to house the video projector and the attachment mechanism; the portable container including a solar cell disposed on an exterior surface of the container, and the solar cell configured to provide electrical power to the video projector; at least one speaker selectively housed in the portable container, and in the solar cell being further configured to provide electrical power to the at least one speaker; and a wireless communication module coupled with the video projector and configured to communicate with the user interface that controls the image data to be projected via the video projector.

(6) According to a second aspect of the present disclosure, a vehicle includes a rear door rotatable between a closed position and an open position. A video projector arrangement includes an attachment mechanism and a video projector. The attachment mechanism selectively couples with the rear door. The video projector is coupled with the attachment mechanism and is configured to project image data in a direction toward the vehicle when the rear door is in the open position.

(7) Embodiments of the second aspect of the disclosure can include any one or a combination of the following features: a handle extending from a vehicle interior surface of the rear door, the attachment mechanism including at least one clip configured to engage the handle; the attachment mechanism including at least one elongate member extending between the video projector and the

at least one clip; and the projector arrangement including a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in the closed position.

(8) According to a third aspect of the present disclosure, a projector arrangement for a vehicle includes an attachment mechanism that selectively couples with the rear door of the vehicle. The attachment mechanism includes a clip and an elongate member. The projector arrangement further includes a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in a closed position. A video projector is coupled with the clip via the elongate member. The video projector is configured to project image data onto the screen when the rear door is in an open position.

(9) Embodiments of the third aspect of the disclosure can include any one or a combination of the following features: at least one speaker, the attachment mechanism, the screen, and the video projector, and the at least one speaker each being configured to be housed in a portable container in a stored position, the portable container including a solar cell configured to charge the video projector and the at least one speaker.

(10) These and other aspects, objects, and features of the present disclosure will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a rear perspective view of a vehicle incorporating the projector arrangement at a rear opening of the vehicle;

(3) FIG. 2A is a perspective view of a portable container according to one aspect of the present disclosure;

(4) FIG. 2B is a partially disassembled view of the container of FIG. 2A;

(5) FIG. 3 is a perspective view of a video projector detached from a rear door of a vehicle according to one aspect of the present disclosure; and

(6) FIG. 4 is a front view of a projector screen coupled with a vehicle adjacent a rear opening of the vehicle according to one aspect of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.

(8) As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

(9) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are

simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(10) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a projector arrangement for a vehicle. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(11) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(12) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(13) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

(14) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

(15) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(16) Referring generally to FIGS. 1-4, the present disclosure relates to a projector arrangement **10** that may enhance a user experience of viewing cinema or the like in an environment around a vehicle **14**. The projector arrangement **10** may be portable via a kit **12** and be configured to selectively couple with a structure of the vehicle **14**. For example, the projector arrangement **10** may be removed from the kit **12** and mounted to the vehicle **14**. Video projected from the projector apparatus may be viewed from an exterior **16** of the vehicle **14** when a rear door of the vehicle **14** is

in an open state, as will be further described herein. The present projector arrangement **10** may allow for less cumbersome steps to be performed by users **18** by, for example, not requiring rearrangement of the vehicle cabin or portions thereof. The portable kit **12** may be stored in the vehicle **14** and removed by the user **18** when the vehicle **14** is stationary, and the portable kit **12** may include the attachments required to provide an enhanced experience. For example, the kit **12** may include various sub-compartments to store the projection equipment, such as audio devices, video devices, stands, panels, screens, and the like, as well as trays or other utensils for holding food items to be consumed by the user **18**. As will be described further herein, the kit **12** may also operate as a charging station or an electrical storage device that is capable of being powered or charged via a solar panel disposed on the kit **12**. In this way, the projector arrangement **10** of the present disclosure may provide for a more enjoyable viewing experience by limiting incompatibility issues with physical features of the vehicle **14**, limiting assembly time, and/or requiring little to no rearrangement of the interior of the vehicle **14**.

(17) Referring now more particularly to FIG. **1**, the vehicle **14** includes a rear door **20** that is pivotable between an open position (as illustrated) and a closed position to selectively cover, or close, a rear opening **22** of the vehicle **14**. The vehicle **14** includes a chassis, or a frame **24**, that includes an overhead panel **26** and one or more upright framing features **28** and lateral framing features **30** that may define the rear opening **22**. As will be described further herein, the supports **28**, **30** may define one or more slots **32** for receiving hanging parts that support at least a portion of the projector arrangement **10**.

(18) Still referring to FIG. **1**, the rear door **20** includes an interior surface **34** that, in the closed position, closes off an interior **36** of the vehicle **14**. The rear door **20** may include one or more handles **38** that may be held by the user **18** to control opening or closing of the rear door **20**. The handles **38** may be a single handle, a pair of handles **38** (as illustrated), or more than a pair of handles **38** that allow the user **18** to shut the rear door **20**. Notches **40** may be formed on the rear door **20** and extend into the interior surface **34** to allow user manipulation of the rear door **20** via the handles **38** and further define an area for the handles **38**. It is contemplated that the handles **38** may be contoured or have a particular curvature or geometry that is configured to mate with at least a portion of the projector arrangement **10**.

(19) For example, the projector arrangement **10** may include an attachment mechanism **42** that selectively couples with the pair of handles **38** of the rear door **20** via a pair of clips **44**. A video projector **46** may be coupled with the attachment mechanism **42** to allow the video projector **46** to project image data in a direction toward the vehicle **14** when the rear door **20** is in the open position. The attachment mechanism **42** may include a pair of elongate members **48**, or supports, that each extend from one of the pair of clips **44** toward the video projector **46**, such that the video projector **46** may be centrally mounted between the clips **44**. The clips **44** and the handles **38** may form a snap-fit connection, such that the clips **44** may be dimensioned slightly more narrowly than the handles **38** to allow the clips **44** to be secured to the handles **38** upon the user **18** applying pressure to the clips **44**. Similarly, the clips **44** may be removed by applying a similar or different pressure outwardly from the rear door **20** to remove the clips **44** from the handles **38**. In this way, the video projector **46** selectively couples with rear door **20** to allow an attached configuration and a detached configuration of the video projector **46**.

(20) With continued reference to FIG. **1**, the projector arrangement **10** may include a projector screen **50** that selectively couples with the vehicle **14** via a plurality of brackets **52** disposed along a periphery **54** of the projector screen **50** when the projector screen **50** is mounted to the vehicle **14**. For example, each bracket **52** may include a first hook **56** and a second hook **58** spaced from and opposing one another. The first hook **56** may engage a wall, or the frame **24**, of the vehicle **14**, and the second hook **58** may engage or couple with the screen **50**. The second hook **58** may be secured with the projector screen **50** via a plurality of holes **60** formed along the periphery **54** of the projector screen **50**. For example, as illustrated, holes **60** may be disposed about four corners of the

projector screen **50**, which may form a generally square shape when coupled with the vehicle **14**. The brackets **52** may allow the projector arrangement **10** to hang from the frame **24** of the vehicle **14** and/or may provide tension to the screen to hold the screen in a generally planar configuration when the projector screen **50** is coupled with the vehicle **14**. For example, brackets **52** along lower parts of the screen **50** may extend laterally or at an oblique angle to hold tension laterally and/or vertically on the screen **50**.

(21) Still referring generally to FIG. **1**, the user **18** of the projector arrangement **10** may control the projector **46** remotely via an interface **66** on a mobile device **64** or an interface associated with the vehicle **14** (e.g., a vehicle human-machine interface (HMI)). For example, the projector arrangement **10** may include a wireless communication module that is configured to communicate with a wireless communication module of the mobile device **64** over a network. The network may be a wireless network, such as the Internet, and may operate with various wired or wireless protocols, such as TCP/IP, 3G, 4G, SMS, Wi-Fi, Bluetooth, Zigbee, or the like to allow the user **18** to select cinema (e.g., movies, shows, and the like) to be streamed on the phone. That video data may be transmitted by the projector **46** toward the projector screen **50** and audio associated with the video data to be presented on an audio communication device, such as vehicle speakers, Bluetooth speakers, or speaker devices further described herein. As will also be further described herein, the projector arrangement **10** may be coupled with an electrical power source when the subassembly (e.g., the projector **46** and attachment mechanism **42**) is attached with the rear door **20** and/or may be powered from a chargeable power device, such as an electrochemical cell (e.g., a battery).

(22) Referring now to FIGS. **2A** and **2B**, the kit **12** may be configured to store various features of the projector arrangement **10** in a container **68** that is portable (e.g., movable). For example, the video projector **46**, the attachment mechanism **42**, the brackets **52**, and the screen **50** may each be storable within various portions of the container **68**. Referring more particularly to FIG. **2A**, the container **68** may include a plurality of sidewalls **72** and a bottom wall (not shown) that couples with the sidewalls **72** to form a cavity **74** within the container **68** for storing various features of the projector arrangement **10**. The container **68** also includes a cover **76** that may be pivotable about one of the sidewalls **72** or may be removable from the sidewalls **72**. In general, the container **68** may be removed from the vehicle **14** and placed outside of the vehicle **14** in a viewing area (e.g., behind the vehicle **14** in the exterior **16**) for viewing a video or other images from image data projected onto the projector screen **50** when the projector arrangement **10** is coupled with the vehicle **14**. Inside the container **68**, for example, a tray **78**, or various sub-compartments of the cavity **74** may be employed for housing food items and/or other contents (e.g., 3D glasses) that may enhance a viewing experience.

(23) Referring now to FIGS. **2A** and **2B** more generally, a solar panel **80** may be provided on an exterior surface **82** of the container **68**. As exemplarily illustrated, the solar panel **80** is coupled with or integrated with a top portion (e.g., cover **76**) of the container **68** in order to provide a greater surface area for capturing solar energy than incorporating the solar panel **80** on the sidewalls **72** of the container **68**. A pair of clasps **84** extend from an opposing pair of sidewalls **72** to allow the user **18** to manipulate the clasps **84** to lift the container **68** and carry the container **68** to/from the vehicle **14**. The lid **76** may define a pair of indentations on an underside of the lid **76** that receive at least one speaker **86** configured to output audio associated with video data transmitted by the projector **46**. These speakers **86** may be secured to the top portion or in the cavity **74** when the projector arrangement **10** is disassembled and in the stored configuration.

(24) With continued reference to FIGS. **2A** and **2B**, the solar panel **80** may include one or more solar cells **88** that are configured to convert solar energy into electrical energy. The electrical energy produced by the one or more solar cells **88** may be converted by an electrical conversion device in the container **68** to a direct-current or alternating current voltage that may be used to charge or power the video projector **46** and/or the speakers **86**. For example, the electrical power may be configured to be output or available for output as a low DC voltage (e.g., 12V, etc.) and/or

115 VAC-120 VAC. Accordingly, an electrical interface may be provided in the container **68** for coupling cords and/or wires between the interface and the video projector **46** or the speakers **86** to charge a battery associated with either or both of the video projector **46** and the speakers **86**. In some examples, the video projector **46** and/or the speakers **86** may be charged wirelessly (via, e.g., wireless induction charging), thereby allowing minimal or reduced involvement of the user **18** with plugging or interconnecting wires from a wall of the container **68** with the subcomponents of the projector arrangement **10**.

(25) Referring now to FIG. 3, the video projector **46** may be configured to mount to the pair of elongate members **48** at a central portion **90** of the subassembly **42**, **46**. In general, each elongate member **48** may extend from the central portion **90** and taper toward ends of the elongate member **48** or an end of the elongate member **48** at the clip **44**. As exemplarily illustrated, each clip **44** may include a pair of opposing latch portions **92** that may deflect away from one another upon pressure being exerted on a back surface of the clip **44** when the clip **44** engages the handle **38**. The deflection of the latch members **92** may cause the latch members **92** to be biased toward one another, such that when the clip **44** engages in the handles **38**, the latch members **92** deflect toward one another within the notch **40** to secure the clips **44** with the handles **38**. In some examples, a coupling of the pair of elongate members **48** with one another may include a hinged portion, such that the elongate members **48** are secured to one another in both the attached and detached configurations. In other examples, the projector **46** mounts to an intermediate plate that is configured to receive each of the elongate members **48**. Additionally, or alternatively, the elongate members **48** and the pair of clips **44** may extend along a radius of curvature or a plurality of radii of curvatures to match or otherwise abut the rear door **20** along a similar contour of the rear door **20**. For example, the elongate members **48** may bow inwardly from the clips **44** toward the projector **46** to reflect the contour of the interior surface **34** and the handles **38** to “hug” the handles **38**, though this example is non-limiting and other configurations, shapes, and contours of the attachment mechanism **42** may be employed. In the example illustrated, the contour may allow the subassembly **42**, **46** to have a relatively low profile, such that the rear opening **22** is not substantially blocked (e.g., blocking user interaction with the rear opening **22**) by the projector **46** when the rear door **20** is in the open position.

(26) Referring now to FIG. 4, the projector screen **50** may be hung or otherwise secured to the walls of the vehicle **14** (e.g., the frame **24**) via the plurality of brackets **52** previously described in relation to FIG. 1. As shown in FIG. 4, the plurality of brackets **52** may be embodied as S— hooks, with the first hook **56** securing to a flange or other surface of the frame **24**. It is contemplated that the first hook **56** of the brackets **52** may engage the apertures as previously described or may engage a ledge, flange, or another surface that protrudes to allow hanging of the bracket **52**. As depicted, the projector screen **50** may cover a substantial portion of the rear opening **22** when in the attached configuration and may also face a vehicle-rearward direction of a passenger compartment of the vehicle **14**, thereby allowing minimal or reduced manipulation of the interior **36** to couple the projector screen **50** with the vehicle **14** and use the projector arrangement **10**.

(27) In general, the projector arrangement **10** of the present disclosure may be generally configured for a plurality of models of vehicles or may be specifically configured for a particular vehicle model, such the interaction of the clips **44** of the handles **38** and/or the contour of the subassembly **42**, **46** may be configured to be flush with the rear door **20**. The projector arrangement **10** may also provide for a relatively simple installation and setup, and may also serve as an all-in-one audio and video connectivity interface. The projector arrangement **10** may allow for a significant user experience of cinema in an off-road environment, and may also allow for or be configured as a rechargeable system that is self-sufficient based on solar power. It is also contemplated that other powering sources for the rear projection apparatus may be incorporated to power and/or charge the video projector **46** and/or the speakers **86** of the present disclosure. Further, the projector arrangement **10** of the present disclosure may limit modification of the interior spaces and

structures of the vehicle **14**.

(28) Modifications of the disclosure will occur to those skilled in the art and to those who make or use the concepts disclosed herein. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims as interpreted according to the principles of patent law, including the doctrine of equivalents.

(29) For purposes of this disclosure, the term “coupled” (in all of its forms: couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature, or may be removable or releasable in nature, unless otherwise stated.

(30) It is also important to note that the construction and arrangement of the elements of the disclosure, as shown in the exemplary embodiments, is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts, or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, and the nature or numeral of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(31) It will be understood that any described processes, or steps within described processes, may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(32) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A projector arrangement for a vehicle, comprising: an attachment mechanism selectively coupling with a rear door of the vehicle; a video projector coupled with the attachment mechanism, the video projector configured to project image data in a direction toward the vehicle when the rear door is in an open position; and a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in a closed position.

2. The projector arrangement of claim 1, wherein the attachment mechanism includes at least one clip configured to engage a handle of the rear door.

3. The projector arrangement of claim 2, wherein the attachment mechanism includes at least one elongate member extending between the video projector and the at least one clip.
4. The projector arrangement of claim 3, wherein the at least one clip includes a first clip corresponding to a first elongate member and a second clip corresponding to a second elongate member, the first and second elongate members laterally spacing the video projector from the handle when the attachment mechanism is coupled with the rear door.
5. The projector arrangement of claim 2, wherein the handle and the clip form a snap-fit connection.
6. The projector arrangement of claim 1, wherein the at least one bracket includes a plurality of brackets, wherein the plurality of brackets are configured to secure the screen in a planar configuration.
7. The projector arrangement of claim 6, wherein each of the plurality of brackets includes a first hook configured to engage a wall of the vehicle.
8. The projector arrangement of claim 7, wherein the screen defines a plurality of holes disposed proximate to a periphery of the screen, and wherein each the plurality of brackets includes a second hook opposite the first hook configured to engage one of the plurality of holes.
9. The projector arrangement of claim 1, further comprising: a portable container configured to house the video projector and the attachment mechanism.
10. The projector arrangement of claim 1, wherein the portable container includes a solar cell disposed on an exterior surface of the container, the solar cell configured to provide electrical power to the video projector.
11. The projector arrangement of claim 10, further comprising at least one speaker selectively housed in the portable container, wherein the solar cell is further configured to provide electrical power to the at least one speaker.
12. The projector arrangement of claim 1, further comprising a wireless communication module coupled with the video projector and configured to communicate with a user interface that controls the image data to be projected via the video projector.
13. The projector arrangement of claim 1, wherein the image data includes at least one of images and video.
14. A vehicle, comprising: a rear door rotatable between a closed position and an open position; a projector arrangement including an attachment mechanism and a video projector, the attachment mechanism selectively coupling with the rear door, wherein the video projector is coupled with the attachment mechanism and is configured to project image data in a direction toward the vehicle when the rear door is in the open position, wherein the attachment mechanism includes at least one clip configured to engage a handle of the rear door.
15. The vehicle of claim 14, further comprising: a handle extending from a vehicle interior surface of the rear door, wherein the attachment mechanism includes at least one clip configured to engage the handle.
16. The vehicle of claim 15, wherein the attachment mechanism includes at least one elongate member extending between the video projector and the at least one clip.
17. The vehicle of claim 15, wherein the handle and the clip form a snap-fit connection.
18. The vehicle of claim 15, wherein the projector arrangement includes a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in the closed position.
19. A projector arrangement for a vehicle, comprising: an attachment mechanism selectively coupling with a rear door of the vehicle, the attachment mechanism comprising: a clip; and an elongate member; a screen and at least one bracket configured to couple the screen with the vehicle proximate to a rear opening of the vehicle closed by the rear door in a closed position; and a video projector coupled with the clip via the elongate member, the video projector configured to project an image onto the screen when the rear door is in an open position.

20. The projector arrangement of claim 19, further comprising: at least one speaker, wherein the attachment mechanism, the screen, the at least one speaker, and the video projector are each configured to be housed in a portable container in a stored position, the portable container including a solar cell configured to charge the video projector and the at least one speaker.
